TopK Sparse Autoencoders Across Three Model Architectures: Dictionary Collapse, Dense-Feature Degeneracy, and the Limits of Activation-Pattern Feature Matching

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Abstract

We train TopK sparse autoencoders (SAEs) at matched sparsity ($k = 128$) and matched dictionary size (16,384) on mid-network residual stream activations from three model families—Llama-3.2-3B, Mistral-7B-v0.3 and Qwen2.5-3B—and report three findings, two of them negative.

First, TopK training without an auxiliary revival loss undergoes an abrupt *dictionary collapse*. In both runs long enough to exhibit it, dead features roughly quadruple between step 2,000 and step 5,000 (Llama: 1,078 → 6,857; Qwen: 2,893 → 11,178) and never recover, because a feature below the top-k threshold on every batch receives exactly zero encoder gradient. Reconstruction quality is unharmed and in fact continues improving throughout, so the collapse is invisible to the metric practitioners usually monitor. Qwen ends at 57.4% of its dictionary permanently dead.

Second, the surviving code is dominated by a small set of dense directions. With $k/D = 0.78\%$, a feature should fire on under 1% of tokens if activation were spread evenly; instead 20–34 features per model fire on more than half of all tokens, and the 200 most frequent features absorb 39–57% of total activation mass. Any procedure that selects features by activation frequency—a common first step in feature labelling and in feature-based classifiers—selects almost entirely from this degenerate head.

Third, cross-architecture feature matching by activation-pattern similarity finds nothing. Under a chunk-permutation null, the best-of-16,384 match similarity between *unrelated* features averages 0.39–0.57 and has a 99th percentile of 0.83–0.94, so the conventional fixed threshold of 0.8 sits at or below the noise floor. With one-to-one reciprocal matching, a closed-triangle requirement and a null-calibrated threshold, the number of features shared across all three models is zero; an earlier uncalibrated analysis of the same checkpoints reported 3,753.

We report a comparison of training objectives as *not conducted*: only TopK was trained, and §5 states plainly which planned experiments do not exist.

1 Introduction

Language models represent concepts in high-dimensional activation spaces that mix thousands of distinct semantic and syntactic features within individual neurons—the phenomenon

Elhage et al. [6] term *superposition*. Sparse autoencoders address this by learning an overcomplete dictionary of directions such that any activation can be reconstructed as a sparse sum over dictionary elements. Bricken et al. [4] applied the approach at scale and found the resulting features highly interpretable.

The practical literature has converged on a small number of quality metrics—variance explained, L_0, dead-feature count—and on reporting them at the end of training. This paper is about what those metrics miss. We train TopK SAEs [7] under matched hyperparameters on three model families and instrument the training process rather than only its endpoint. Two pathologies emerge that a final-checkpoint report would not surface, and a third result—an attempt to match features across the three models—fails in a way that we think is instructive.

Why these three models. Llama-3.2-3B, Mistral-7B-v0.3 and Qwen2.5-3B come from three organizations, have different hidden dimensions (3,072 / 4,096 / 2,048), different attention schemes, and no publicly shared training data. Holding dictionary size fixed at 16,384 across them means the expansion ratio varies (5.33× / 4.0× / 8.0×), which turns out to matter: the model with the largest expansion ratio collapses hardest.

Contributions.

1. **Dictionary collapse is abrupt, mid-training, and invisible to FVE.** We characterize a sharp transition between steps 2,000 and 5,000 in which dead features quadruple while reconstruction quality continues to improve (§4.1).
2. **Quantification of dense-feature degeneracy.** We measure how much of the sparse code's activation mass is carried by its most frequent features, and show the concentration is severe enough to invalidate frequency-based feature selection (§4.2).
3. **A permutation null for cross-model feature matching, and a negative result.** We show that the standard protocol—fixed cosine threshold, nearest-neighbour matching—produces thousands of spurious matches, and that nothing survives proper calibration (§4.4).
4. **A held-out generalization measurement** for the one SAE that has unseen data, and an explicit statement of why the other two do not (§4.3).

2 Related Work

2.1 Superposition and Dictionary Learning

Elhage et al. [6] established the theoretical basis for SAE-based feature extraction: a transformer with n neurons can represent $O(n^2)$ nearly orthogonal feature directions if those features are sparse. Dictionary learning recovers them by sparse factorization: given activations $X \in \mathbb{R}^{n\times d}$, find $D \in \mathbb{R}^{m\times d}$ (with $m \gg d$) and sparse $C \in \mathbb{R}^{n\times m}$ such that $X \approx CD$ and $\|c_i\|_0 \ll m$ for each row.

2.2 SAE Architecture Variants

Standard L1 SAE. Bricken et al. [4], Cunningham et al. [5] train an encoder–decoder pair to minimize

$$\mathcal{L} = \|x - \hat{x}\|_2^2 + \lambda \|z\|_1, \tag{1}$$

where $\hat{x} = W_{\text{dec}}\,\text{ReLU}(W_{\text{enc}}x + b_{\text{enc}}) + b_{\text{dec}}$. The L1 penalty encourages sparsity through a continuous relaxation.

TopK SAE. Gao et al. [7] replace the L1 penalty with a hard constraint:

$$z^{\text{TopK}} = \text{TopK}(\text{ReLU}(W_{\text{enc}}x + b_{\text{enc}})), \tag{2}$$

guaranteeing exactly k nonzero activations per input. Critically for this paper, Gao et al. [7] pair TopK with an *auxiliary loss* that periodically resurrects dead feature directions. Our implementation omits it, and §4.1 is in effect a measurement of what that omission costs.

Gated and JumpReLU SAEs. Rajamanoharan et al. [8] separate feature detection from magnitude estimation via a gating network; raj [1] use a discontinuous activation with a learned threshold. We did not train either variant (§5).

2.3 Evaluation Methodologies

Prior evaluation approaches include linear probing [2, 3, 5], human interpretability ratings via the top-k activating contexts protocol [4], and activation steering fidelity [9, 10]. This paper uses none of them; our evaluation is confined to reconstruction quality, dictionary utilization, and cross-model feature correspondence.

3 Methods

3.1 Models, Layers, and Activation Collection

SAEs are trained on residual stream activations at the 50% depth midpoint of each model: **Llama-3.2-3B** (layer 14 of 28, d = 3,072), **Mistral-7B-v0.3** (layer 16 of 32, d = 4,096), and **Qwen2.5-3B** (layer 18 of 36, d = 2,048). Activations are collected from 500,000 tokens of WikiText-103 (`wikitext-103-raw-v1`, train split), streamed in the same order for every model, and stored as float16.

Mistral activations were extracted from 4-bit quantized weights (`mlx-community/Mistral-7B-v0.3-4bit`); the bf16 base model requires authentication we did not have. This is a confound we cannot rule out and flag wherever Mistral results appear.

3.2 SAE Configuration

All three SAEs are TopK with $k = 128$, dictionary size D = 16,384, peak learning rate 10^{-4} with cosine annealing, batch size 2,048 tokens, and seed 42. Because D is held fixed while d varies, the expansion ratio differs by model: 5.33× (Llama), 4.0× (Mistral), 8.0× (Qwen). The encoder is $z = \text{TopK}(\text{ReLU}(W_{\text{enc}}(x - b_{\text{dec}}) + b_{\text{enc}}))$ and the decoder $\hat{x} = W_{\text{dec}}z + b_{\text{dec}}$, with decoder columns unit-normalized. The loss is plain MSE; there is no auxiliary dead-feature loss and no re-initialization scheme.

Table 1: The three training runs. Only Qwen reached the step target. “Epochs” is the number of passes over the source token set implied by steps × batch.

Model	Layer	d	Expansion	Source tokens	Steps	Epochs
Llama-3.2-3B	14	3,072	5.33×	500,000	13,000	53
Mistral-7B‡	16	4,096	4.0×	50,000	1,000	41
Qwen2.5-3B	18	2,048	8.0×	500,000	50,000	205

‡ 4-bit quantized source model.

Training runs differ in length, and not by design (Table 1). Only the Qwen run reached its 50,000-step target. This unevenness limits cross-model comparison throughout, and we do not attribute differences between models to architecture where training length is a plausible alternative explanation.

3.3 Reconstruction Metrics

We report fraction of variance explained, $\mathrm{FVE} = 1 - \mathrm{Var}(x - \hat{x}) / \mathrm{Var}(x)$, and MSE. MSE is activation-scale-dependent and is not comparable across models with different hidden dimensions and activation norms; FVE is.

An important measurement detail: during training, FVE is computed on the current batch. This is a noisy estimator, and reporting the final-step value—the natural thing to do—samples once from a wide distribution. Across the last ten logged steps the batch FVE spans 0.013 (Llama), 0.021 (Qwen) and 0.364 (Mistral). All FVE figures in §4 are instead computed over the full 500,000-token corpus.

3.4 Cross-Architecture Feature Matching

Features are compared across models by *functional* similarity: the correspondence of their activation patterns over a shared corpus. For each model we encode 500,000 tokens of WikiText-103, partition them into $N = 1{,}000$ non-overlapping chunks of 500 tokens, and average post-TopK feature activations within each chunk. Column j of the resulting $N \times D$ matrix is a 1,000-dimensional *fingerprint* of feature j. Fingerprints are centred before normalization, so the similarity statistic is a Pearson correlation across chunks rather than a cosine between non-negative vectors.

Three procedural choices distinguish this from the conventional protocol, and §4.4 shows each is necessary:

1. **Support filter.** Features active in fewer than 5% of chunks are excluded. A feature active in a handful of chunks correlates near 1 with any other feature active in the same chunks, by construction rather than by shared function.

2. **Reciprocal matching.** A match requires features i and j to be *mutual* nearest neighbours. This is one-to-one by construction. Taking the union of both directions' nearest neighbours—the common alternative—permits one feature to be the recorded partner of arbitrarily many, and permits match counts exceeding the dictionary size.

Table 2: Dead-feature count (of 16,384) over training. Both runs long enough to show it undergo a sharp transition between steps 2,000 and 5,000. FVE is the batch estimate at that step; note that it does not degrade across the transition. Data: `data/sae-runs/*/training.jsonl`.

	Llama-3.2-3B		Qwen2.5-3B		Mistral-7B	
Step	Dead	FVE	Dead	FVE	Dead	FVE
1	1,381	−0.109	3,220	−0.096	748	−0.088
200	1,245	0.624	3,091	0.743	468	0.829
1,000	1,145	0.955	2,975	0.948	339	0.965
2,000	1,078	0.986	2,893	0.982	—	—
5,000	**6,857**	0.775	**11,178**	0.841	—	—
10,000	6,273	0.992	9,622	0.989	—	—
13,000	5,278	0.984	9,538	0.979	—	—
50,000	—	—	9,401	0.984	—	—

3. **Null-calibrated threshold.** τ is set to the 99th percentile of a permutation null in which chunk order is shuffled in one model, destroying cross-model correspondence while preserving each feature's marginal activation profile (20 replicates).

A feature triple is *three-way universal* only if all three pairwise edges independently satisfy these criteria—a closed triangle, not a chain through an intermediate model.

4 Results

4.1 Dictionary Collapse

Table 2 tracks dead features—those with no activation over a rolling 5,000-step window—through training.

Three features of this are worth stating explicitly.

The collapse is abrupt. Dead features are roughly flat and slowly *declining* through step 2,000 in both long runs, then jump by a factor of 6.4× (Llama) and 3.9× (Qwen) by step 5,000. This is not gradual attrition. The mechanism is a ratchet: under a hard top-k constraint, a feature that falls outside the top k on every batch in the window receives an encoder gradient of exactly zero, so it cannot be pushed back into contention by training. Once the learning rate has annealed enough that the surviving features stop reshuffling, the set of live features is frozen.

Reconstruction quality does not warn you. FVE at step 5,000 dips (0.775, 0.841) but this is batch noise, not signal: by step 10,000 both runs are at 0.99, higher than before the collapse, and they stay there. A practitioner monitoring FVE would see a healthy, monotonically improving run. The dictionary lost 40–60% of its capacity and the loss curve did not notice, because the surviving features simply absorbed the reconstruction work.

Table 3: Concentration of activation mass, measured over the full 500,000-token corpus. "Top-200 mass" is the share of all feature activations accounted for by the 200 most frequently firing features—1.2% of the dictionary. Data: `data/sae-analysis/corrections.json`.

Model	Uniform expectation	Features firing >50% of tokens	Top-200 mass
Llama-3.2-3B	0.78%	34	40.3%
Qwen2.5-3B	0.78%	20	38.6%
Mistral-7B	0.78%	34	57.2%

Higher expansion collapses harder. Qwen (8.0× expansion) ends with 57.4% of its dictionary dead against Llama's 20.2% (5.33×), measured over the full corpus. This is the expected direction—a larger dictionary relative to d has more features competing for the same k slots—but with two long runs we cannot separate it from the confound that Qwen also trained nearly four times as long. Mistral (4.0×, 1,000 steps) shows 0.1% dead, which reflects only that it stopped before the transition.

4.2 Dense-Feature Degeneracy

Sparsity of the code does not imply that the code is evenly used. With $k = 128$ of $D = 16{,}384$, a feature would fire on 0.78% of tokens if activation were spread uniformly. Table 3 shows the actual distribution.

In every model, 1.2% of the dictionary carries roughly two-fifths of the sparse code, and a few dozen features fire on the majority of all tokens. A feature firing on 97% of tokens—the maximum we observe in both Llama and Qwen—is not a feature in any useful sense; it is closer to a learned bias absorbed into the dictionary.

Consequence for feature selection. The standard first step in feature labelling is to take the top-N features by activation frequency and inspect their maximally activating contexts. Table 3 says that this procedure draws almost exclusively from the degenerate head: of the 200 most frequent Llama features, 34 fire on more than half of all tokens. In separate work we built a feature-based safety classifier on exactly this selection and obtained ROC-AUC 0.564 with a 95% confidence interval spanning chance. Selecting by *differential* activation between contrastive corpora, and imposing a density ceiling, are both necessary; neither is standard practice.

4.3 Held-Out Reconstruction

The Llama and Qwen SAEs each made 53 and 205 passes over their entire activation dump. There is no unseen data for them, and at that level of repetition an SAE can in principle memorize the corpus. We report their FVE as the best available characterization, not as evidence of generalization.

Mistral is the exception, by accident: trained on only the first 50,000 tokens, it leaves 450,000 unseen. Its held-out FVE is 0.9252 against an in-sample 0.9273—a gap of 0.2 percentage points. A 1,000-step SAE has many problems, but overfitting its training corpus

Table 4: Reconstruction over the full corpus. Only Mistral has unseen data, because it was trained on the first 50,000 tokens of a 500,000-token dump; its held-out row covers tokens 50k–500k. Dead-feature counts here are features that fire nowhere in the corpus, a stricter criterion than the rolling window of Table 2.

Model	Split	MSE	FVE	Dead features	
Llama-3.2-3B	in-sample	0.00788	0.9865	3,309 (20.2%)	53 epochs
Qwen2.5-3B	in-sample	0.27532	0.9839	9,398 (57.4%)	205 epochs
Mistral-7B	in-sample	0.00415	0.9273	24 (0.1%)	41 epochs
Mistral-7B	**held-out**	0.00426	**0.9252**	26 (0.2%)	unseen

Table 5: Matchable feature inventory over the 500,000-token corpus. "Dead" features fire nowhere; "low support" features fire in fewer than 50 of 1,000 chunks. Only the last column enters the matching analysis.

Model	Dictionary	Dead	Low support	**Matchable**
Llama-3.2-3B	16,384	3,309	4,771	**8,304** (50.7%)
Qwen2.5-3B	16,384	9,398	22	**6,964** (42.5%)
Mistral-7B	16,384	24	10,881	**5,479** (33.4%)

is not among them. This is weak evidence that the other two figures are not purely memorization; it comes from the least-trained model in the set and should not carry much weight. Reserving a held-out split before training is the cheapest fix available to this work.

4.4 Cross-Architecture Feature Matching

Between a third and a half of each dictionary is usable (Table 5), and the two failure modes are the ones diagnosed above: Qwen is half dead from collapse, Mistral is overwhelmingly low-support because its features never specialized.

The null. Table 6 reports the chunk-permutation null. The number to attend to is the mean: two features from different models with no functional relationship, matched reciprocally over a 1,000-dimensional fingerprint, correlate at 0.39–0.57 on average.

A fixed threshold of $\tau = 0.8$—the conventional choice, and the one used in an earlier analysis of these same checkpoints—is at or below that noise floor for two of three pairs. The problem is structural rather than a matter of picking a better number: a match is the best of $\sim$5,000–8,000 candidates, and the expectation of an extreme-value statistic grows with the size of the search and shrinks with fingerprint dimensionality. At a lower resolution we also tested (100 chunks over 50,000 tokens, a common configuration) the null 99th percentile reaches 0.98–0.995, leaving the statistic with no discriminative power whatsoever.

Results. Table 7 gives the counts.

No feature matches across all three models. The closed-triangle count is zero against a null expectation of 0.15 per replicate. Pairwise, 10–22 pairs survive per model pair against 5–7 expected by chance—an enrichment of 2–3×, or on the order of 5–15 genuine pairs out

Table 6: Chunk-permutation null, 20 replicates. τ is set to the null 99th percentile. "Expected FP" is the number of matches a shuffled control produces above τ. Data: `data/sae-analysis/matching-v2/matching-report.json`.

Pair	Null mean	Null p99 ($= \tau$)	Null matches/replicate	Expected FP
Llama–Qwen	0.574	0.939	700	7.0
Mistral–Qwen	0.393	0.832	486	4.9
Llama–Mistral	0.464	0.894	557	5.6

Table 7: Cross-architecture matches under the corrected protocol, against the uncalibrated protocol applied to the same checkpoints. Enrichment is observed matches over expected false positives.

Pair	Uncalibrated	**Corrected**	Expected FP	Enrichment
Llama–Qwen	5,923	**22**	7.0	3.1×
Mistral–Qwen	8,099	**10**	4.9	2.0×
Llama–Mistral	12,174	**12**	5.6	2.1×
Three-way universal	**3,753**	**0**	0.15	—

of 5,000–8,000 matchable features (0.1–0.3%). At these counts Poisson uncertainty alone spans a factor of two, and we do not think the pairwise numbers support a quantitative claim.

The three corrections compound and none is individually sufficient. Reciprocal matching alone reduces the Llama–Mistral count from 12,174 to 40 at the *same* threshold, because the original figure counted a handful of Mistral features many times over. The closed-triangle requirement removes chained triples—in the uncalibrated run, 67% of nominally universal triples had no verified Llama–Mistral edge, and the 3,753 triples collapsed onto 644 distinct Qwen and 341 distinct Mistral features. Null calibration removes the remainder.

What this does and does not show. It does not show that these three models fail to share representational structure. Two of the three SAEs are undertrained, all three have degraded dictionaries, and the fingerprint statistic—mean activation over 500-token chunks—discards within-chunk structure, so a feature responding to a rare token is indistinguishable from one responding to a chunk's topic. The negative result is confounded with the quality of the instruments.

What it does show is methodological and transfers beyond this dataset: cross-model feature matching by activation-pattern similarity requires a permutation null. Without one, the search over a large dictionary manufactures high similarities from nothing, and a threshold that reads as stringent can sit below the noise floor.

5 Planned Experiments Not Conducted

An earlier version of this document was framed as a three-way comparison of SAE training objectives across three evaluation metrics, and stated results for that comparison in its abstract. Those results did not exist. We list here what was planned and not run, so that no reader has to reconstruct it.

- **L1, Gated and JumpReLU SAEs were never trained.** Only TopK exists. Every claim about how objectives compare—including the claim that reconstruction quality mediates probe accuracy—has no data behind it.
- **No linear probing study.** No probes were trained on SAE features, so there is no probe-accuracy result and no mediation analysis.
- **No human interpretability study.** No annotators were recruited, no features were rated, and no inter-rater agreement was computed. The ethics statement in the earlier draft described compensation and institutional approval for a study that did not take place.
- **No steering-fidelity evaluation of SAE features.** Separate steering experiments exist in this workspace, but they use contrastive mean-difference vectors, not SAE decoder directions, and do not bear on the objective comparison.
- **No sparsity-level ablation.** All runs use $k = 128$; the planned $L_0 \in \{16, 32, 64\}$ sweep was not run.
- **GPT-2 Small and Pythia-1.4B were never used.** The earlier draft named them as the model set. No SAE was trained on either.

The objective comparison remains, in our view, the right experiment—the fragmentation in the literature it was meant to address is real. It is simply not this paper.

6 Discussion

Monitor dictionary utilization, not just reconstruction. The collapse in §4.1 is invisible in FVE and in the loss curve. It is visible only in a dead-feature count, and only if that count is logged during training rather than read off the final checkpoint. Since the auxiliary loss of Gao et al. [7] is designed precisely to prevent it, the practical recommendation is simply to use it—but the measurement matters independently, because a run using the auxiliary loss still needs verification that it worked.

Report reconstruction over a corpus, not a batch. Batch FVE at the final step spans 0.36 across adjacent logged steps in our shortest run and 0.02 in our longest. A single reading is a draw from that distribution. Full-corpus evaluation costs one forward pass and removes the ambiguity entirely.

Expansion ratio is a hyperparameter, not a free choice. Holding dictionary size constant across models with different hidden dimensions—a natural way to make a comparison "controlled"—silently varies expansion ratio, and the model with the highest ratio collapsed hardest. Whether expansion ratio drives collapse or merely correlates with it here, we cannot say with two long runs.

Frequency-based feature selection selects the wrong features. This follows directly from §4.2 and is, we suspect, the most transferable practical finding. It affects any pipeline whose first step is "take the most active features."

6.1 Limitations

Only one objective. Everything reported here is about TopK SAEs without an auxiliary loss. The collapse dynamics in particular are specific to hard-sparsity training; L1 SAEs fail differently.

Uneven and insufficient training. One of three runs reached its target. The Mistral SAE trained for 1,000 steps on a corpus ten times smaller than the others. Cross-model differences are confounded with training length throughout, and we have avoided attributing any difference to architecture.

No held-out data for two of three models. See §4.3.

Quantization confound. Mistral activations come from 4-bit weights.

Single corpus, single seed. All results use WikiText-103 and seed 42. There is one run per condition, so we report no variance across seeds and cannot say whether the collapse step is stable.

No semantic labelling. We never establish what any feature represents. The matching analysis asks whether activation patterns correspond, not whether the corresponding features mean the same thing—and with zero three-way matches, the question does not arise.

7 Conclusion

We trained TopK SAEs at matched sparsity and dictionary size on three model families and instrumented the training rather than only its endpoint. Without an auxiliary revival loss, the dictionary collapses abruptly between steps 2,000 and 5,000, losing 40–60% of its capacity while reconstruction quality continues to improve—a failure mode that the usual metrics conceal. What survives is dominated by dense directions: 1.2% of the dictionary carries roughly 40% of the sparse code, which invalidates frequency-based feature selection.

Cross-architecture feature matching finds nothing that survives a permutation null. The conventional protocol applied to these same checkpoints reports 3,753 shared features; the corrected count is zero. The gap is entirely attributable to three procedural choices—many-to-one matching, chained rather than closed triples, and an uncalibrated threshold—each of which is easy to make and none of which is visible in the output.

We report the objective comparison this paper was originally framed around as not conducted.

Reproducibility Statement

All SAE training code, activation collection scripts, training logs, checkpoints, and analysis scripts are available at `https://github.com/jmelton-research/sae-crossarch`. All runs used seed 42. Training was conducted on Apple Silicon using the MLX framework. The corrected matching analysis is `data/sae-analysis/cross_arch_matching_v2.py` and the superseded version is retained as `cross_arch_matching.py` for comparison; reconstruction and null statistics are regenerated by `data/sae-analysis/recompute_corrections.py`.

Ethics Statement

This work involves no human subjects and no personal data; all activations are derived from WikiText-103, a public corpus of Wikipedia text. An earlier draft of this document contained an ethics statement describing annotator compensation and institutional approval for a human interpretability study that was never conducted; that statement has been removed.

SAEs and the feature-matching methods studied here have applications in interpretability and safety research and potential for misuse in targeted manipulation of model behaviour. Our results are negative and release no feature directions of either kind.

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